

Highlights

- A compact recurrent SOC estimator is benchmarked against three conventional classes.
- Cell-disjoint tests separate nominal accuracy, robustness, and recovery.
- Structured pruning reduces the GRU hidden width from 96 to 67 units.
- Continuous state cuts neural latency from 724 ms to 0.425 ms on STM32.
- Periodic state resets expose large transients despite low resource demand.